

Enhancing Data Ingestion and Integration: Success with Talend Data Hub

A leading home services provider partnered with Artha Solutions to optimize data flow. By deploying a Talend Data Integration Hub, they streamlined data ingestion from 300+ utility partners. This enhanced data validation, automation, and scalability, improving data integrity, operational efficiency, and partner onboarding, leading to faster service delivery.

Outcomes

50%

Reduction in manual data processing time, improving overall efficiency.

300+

Utility partners' data seamlessly ingested, ensuring real-time data availability.

35%

Improvement in data accuracy, reducing errors in partner transactions.

60%

Increase in automation, streamlining partner onboarding and data validation.

25%

Faster issue resolution, enabling more reliable and efficient customer service.



Challenges

As a growing home services company managing thousands of daily transactions, the provider faced several challenges in data ingestion and processing:

Data Validation and Integrity

Ensuring real-time validation of incoming partner data, applying custom rules to eliminate errors before processing.

Automation of Ingestion Processes

Reducing manual intervention by dynamically adjusting to unique data structures from different partners.

Error Handling and Data Stewardship

Tracking and resolving data errors efficiently to ensure smooth business operations.

Handling Multiple Data Sources and Formats

Managing data from 300+ utility partners, including file-based inputs (SFTP uploads) and database connections.



Solutions Provided



Metadata-Driven Processing for Automation

Developed a metadata-driven ingestion approach that dynamically adjusts to diverse data structures, minimizing manual intervention, reducing processing errors, and enhancing scalability.



Rule-Based Data Validation and Stewardship

Integrated custom validation logic to ensure only high-quality data is processed, while enabling real-time validation feedback with pass/fail status, accelerating error resolution and data integrity.



Scalable, Automated Data Ingestion Platform

Designed and deployed a Talend Data Integration Hub (IDH) to process data from 300+ utility partners, incorporating Managed File Transfer (MFT) via SFTP for seamless integration with the ERP system.



Historical Data Storage and Compliance Auditing

Implemented persistent data storage for historical tracking, auditing, and compliance reporting, while developing intermediate storage to enable seamless integration with the Ensura Web Services API.